A Human-Like Agent For Screen-Aware Task Automation

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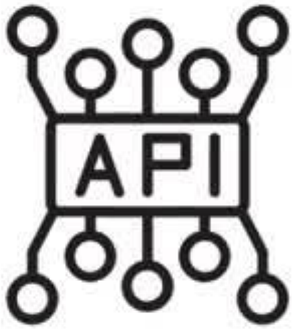

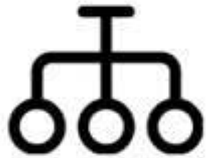

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Title of the individual component: Visual Perception

Specialization: Software Engineering

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IT22099518 | Hemapriya H.A.N.S | 25-26J-477

Individual Title: Visual Perception

💡 Proven Gap / Creative Solution

- Current automation tools (Selenium, Appium, UIAutomator) require internal access (DOM, APIs, app-level hooks) → unsuitable for black-box testing.
- No reliable solution exists for external, human-like screen observation.
- **Creative solution:** external camera + computer vision + OCR + VLM → enables cross-platform, non-intrusive, black-box test automation.

🔍 Knowledge Gap / Problem Definition

- Tools lack ability to perceive screens visually like humans → fail in legacy or restricted environments.
- Existing OCR/UI detection models struggle with real-time noisy inputs (glare, blur, distortions).

⚠️ Shortcomings of Existing Systems & Proposed Solution

Shortcomings	Proposed Solution
• require internal access	• no need for system-level or app-level access.
• Limited to specific platforms	• Cross-platform black-box automation
• OCR/UI detection in existing tools not optimized for real-time noisy inputs	• Preprocessing + Computer Vision + OCR for robust recognition.

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Individual Title: Visual Perception

Key Domains of Knowledge

- Computer Vision – OpenCV, deep learning models for object/element detection.
- OCR – Tesseract, EasyOCR, PaddleOCR for text extraction across variable fonts/colors.
- VLM (Vision-Language Models) – CogAgent, ShowUI, LLaVA for grounding perception with language reasoning.

Latest Technologies

- Real-time external camera feed capture.
- Preprocessing pipelines for clarity improvement.
- Region of Interest (ROI)-based change detection to minimize processing overhead.
- VLM integration for semantic understanding of screen states.

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Individual Title: Visual Perception

Evaluation & Validation

- Accuracy of UI element detection ≥ 90%.
- OCR success rate across varied interfaces.
- Latency Testing → Evaluate frame-to-JSON conversion time (goal < 500ms).
- Robustness → Test under different screen resolutions, lighting conditions, and angles.

Data & Ethical Clearance

- Data Type → Only visual data (screenshots, screen recordings) captured via external camera; no personal data collected from system memory.
- Anonymization → Sensitive text (e.g., usernames, passwords) blurred or excluded in dataset preparation.
- Data Storage → Images processed in-memory; no long-term storage unless explicitly allowed.
- Ethical Approval → Conforms to software testing ethics no unauthorized access to internal APIs or private system data.

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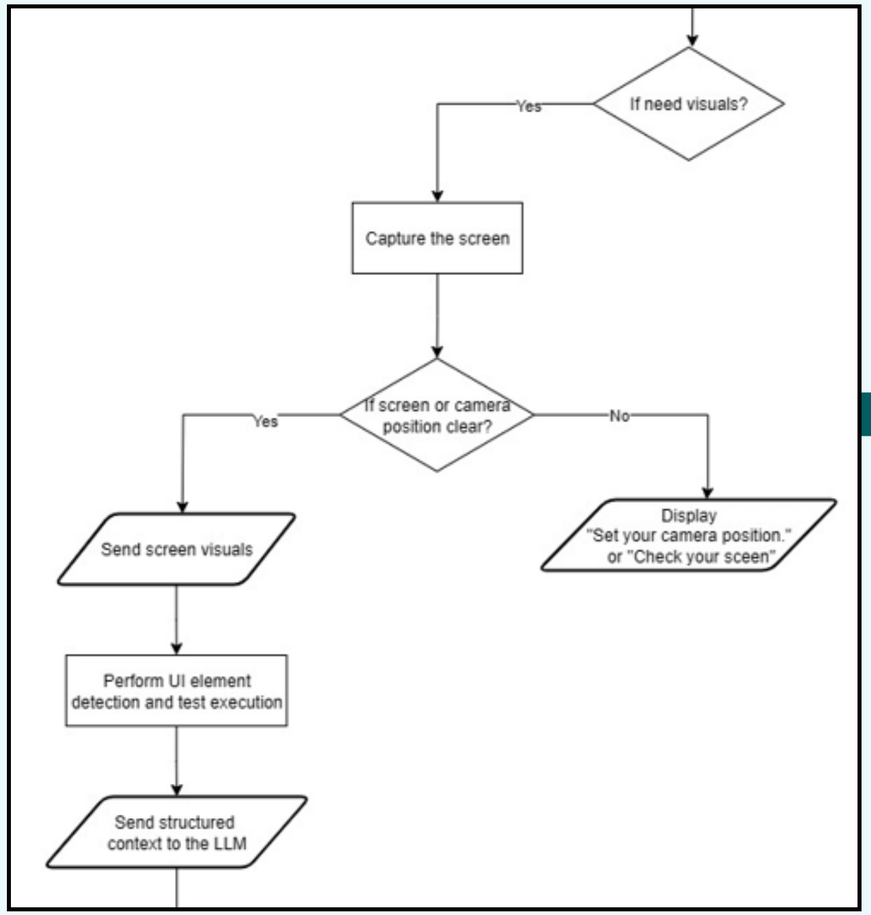
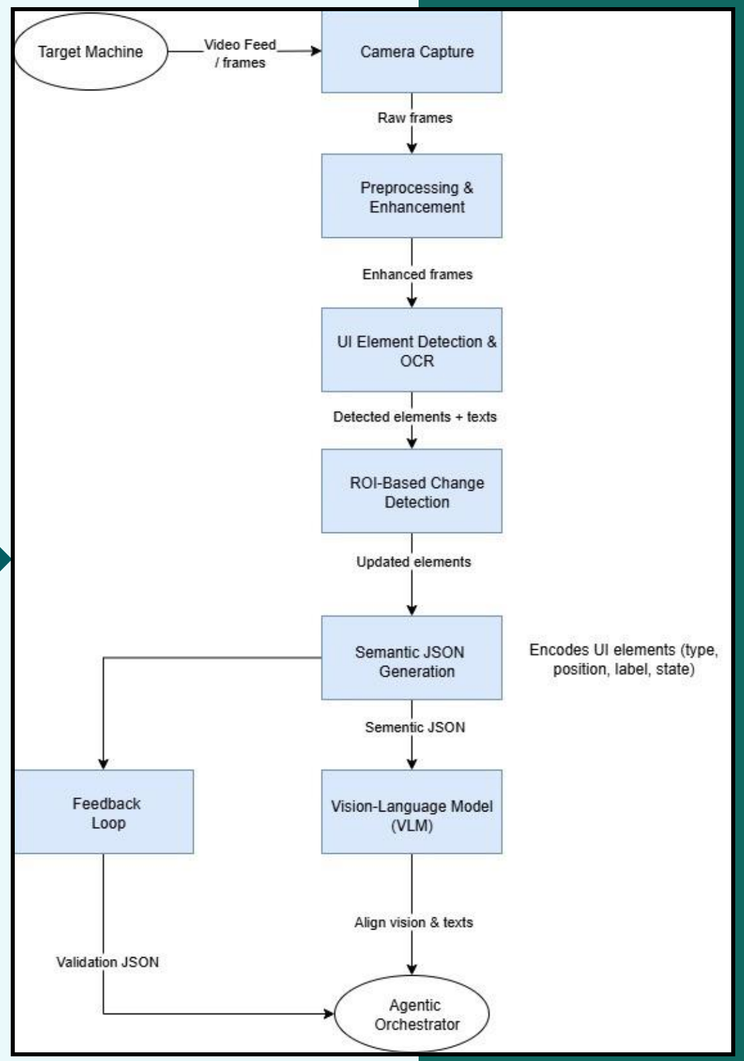
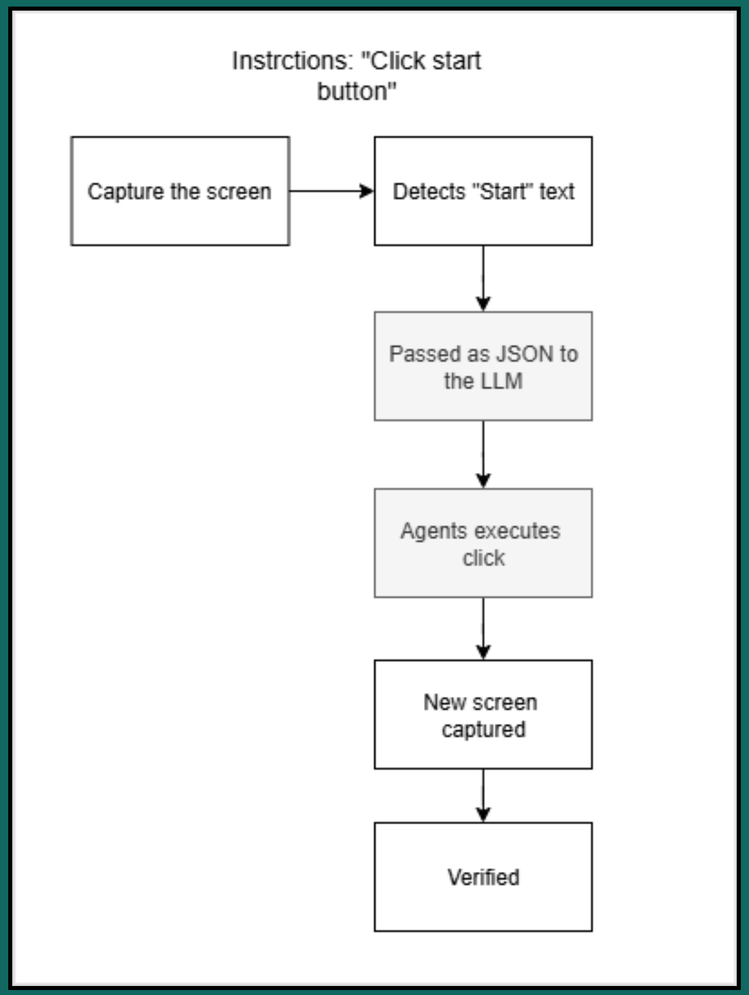
Individual Title: Visual Perception

System Overview





Real-World Usage Scenario



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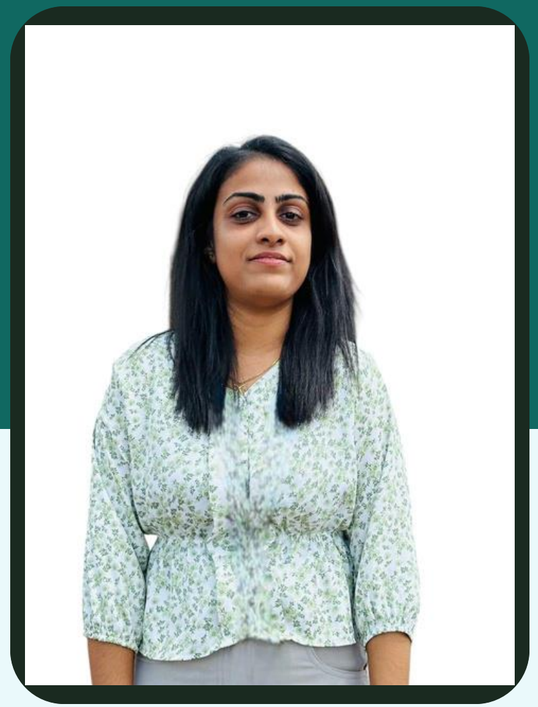
IT22099518 | Hemapriya H.A.N.S | 25-26J-477

Title of the individual component: LLM & Prompt Engineering

Specialization: Software Engineering



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Individual Title: LLM & Prompt Engineering

💡 Proven Gap / Creative Solution

- Most tools (like Selenium, Appium) need pre-defined scripts.
- No natural language support – testers must write scripts.
- **Creative solution:** Our solution uses natural language instructions, LLM-generated actions, and adapts dynamically without manual scripting.

🔍 Knowledge Gap / Problem Definition

- Existing automation tools (e.g., Selenium, Appium) require scripted test cases, limiting adaptability.
- No support for natural language instructions.
- Failures occur when UI changes unexpectedly.
- **Problem Definition:** UI automation is rigid, script-based, and breaks with interface changes.

⚠️ Shortcomings of Existing Systems & Proposed Solution

Shortcomings	Proposed Solution
• Existing tools need scripts.	• No manual scripting needed.
• Lack natural language support.	• Accepts natural language instructions.
• Struggle with changes.	• Adapts to dynamic UI changes.
• Require manual prompt setup.	• Prompts created using prompt engineering.

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Individual Title: LLM & Prompt Engineering

Key Domains of Knowledge

- NLP – process tester instructions
- LLMs – generate step-by-step actions
- Prompt Engineering – create context-aware prompts
- UI Automation – handle UI elements and events
- AI Context Awareness – manage dynamic UI changes

Latest Technologies

- Large Language Models (LLMs) – GPT-4, LLaMA
- Prompt Engineering Tools – LangChain, PromptLayer

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Individual Title: LLM & Prompt Engineering

Evaluation & Validation

- Accuracy of Action Plans: LLM steps correctly perform intended UI tasks.
- Natural Language Understanding: Instructions and synonyms correctly interpreted.
- Robustness: Handles dynamic UI changes and unexpected inputs.
- Efficiency: Reduces manual scripting and test time.

Data & Ethical Clearance

- Data Sources: Use anonymized or synthetic UI data for testing.
- Privacy: Ensure privacy by excluding sensitive user information.
- Ethical AI Use: Use AI responsibly and fairly.
- Compliance: Follow institutional rules and policies.

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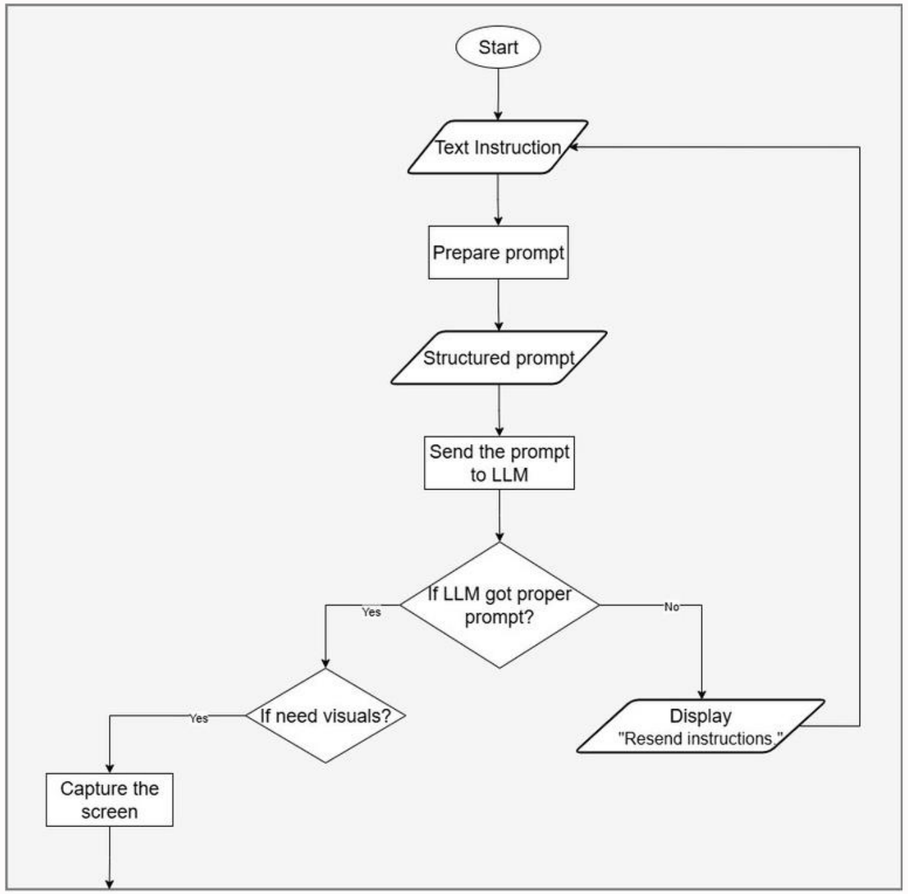
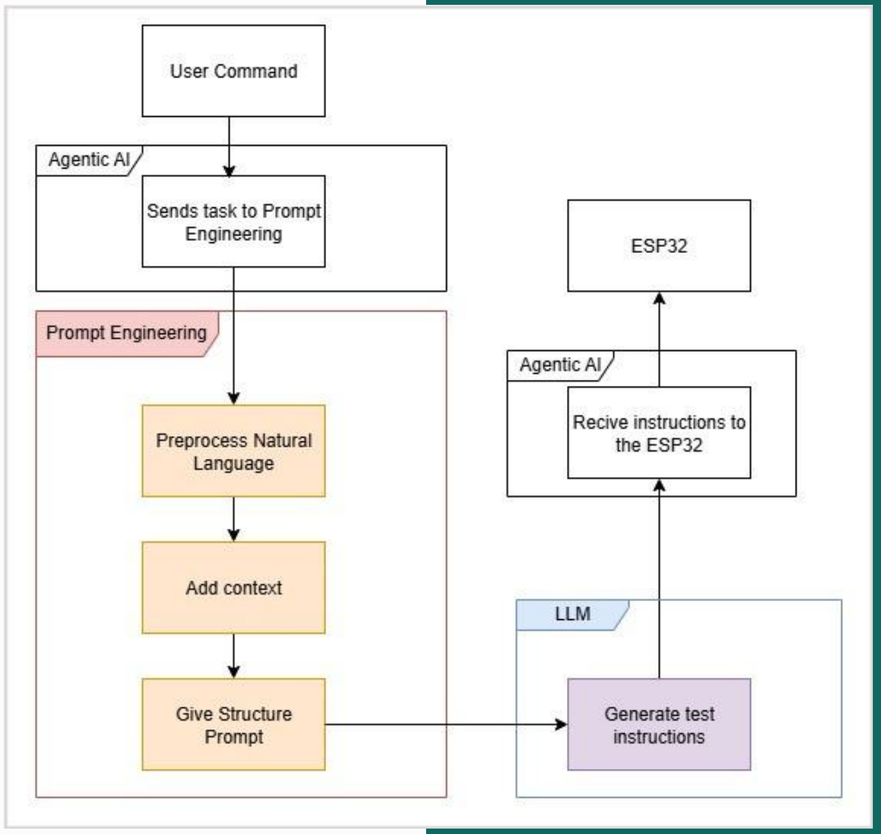
Individual Title: LLM & Prompt Engineering

System Overview

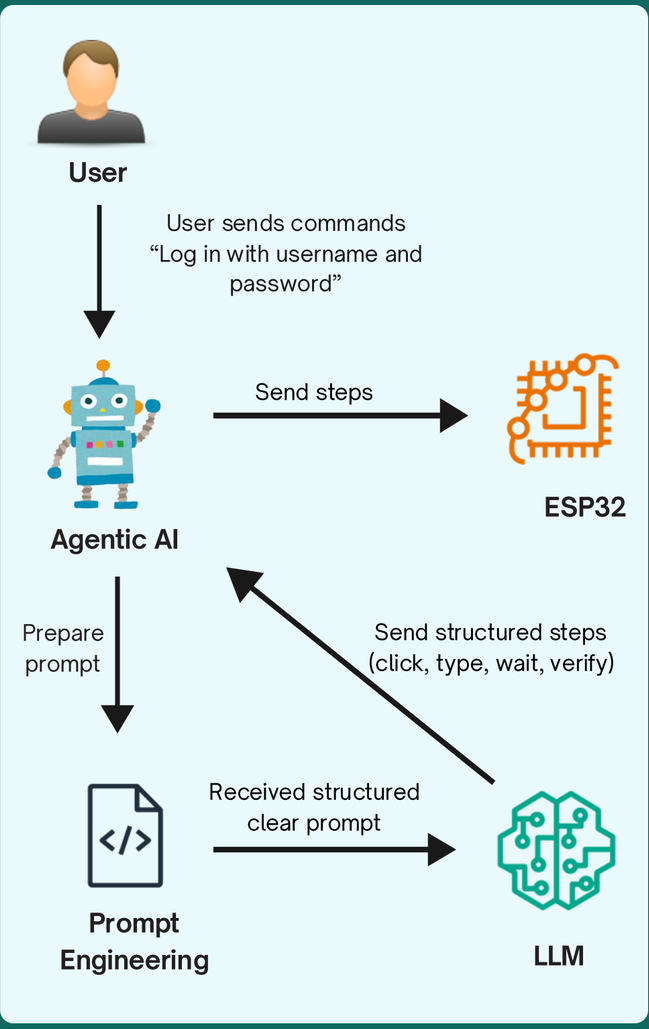
The system converts user instructions into executable UI actions, adapts dynamically using visual feedback, and enables platform-independent, black-box automation without internal system access.





Real-World Usage Scenario

- *Example: Login to Dashboard with username and password*



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Title of the individual component: HID Actuation

Specialization: Software Engineering

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IT22167378 | Hasintha H.I.G.A | 25-26J-477

Individual Title: HID Actuation

💡 Proven Gap / Creative Solution

- Existing HID emulators (e.g., Rubber Ducky, basic Arduino HID) are static, pre-scripted and lack adaptability.
- They cannot provide feedback on execution.

Creative solution: an intelligent HID actuation module that:

- Works in real time, driven by Agentic AI commands.
- Provides bidirectional communication (execution → feedback → retry).

🔍 Knowledge Gap / Problem Definition

- Most existing solutions rely on API access or the ability to inject code into the target application.
- Such approaches fail for legacy, black-box, or restricted software, where internal access is impossible
- How can an ESP32-based HID emulation system reliably reproduce human-like keyboard and mouse interactions?

⚠️ Shortcomings of Existing Systems & Proposed Solution

Shortcomings	Proposed Solution
• Static execution, no feedback loop	• Dynamic & adaptive
• Limited reliability in dynamic, real-world testing.	• Execution feedback loop
• fail in restricted environments.	• Lightweight & undetectable
• Require manual prompt setup.	• No manual scripting needed

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Individual Title: HID Actuation

Key Domains of Knowledge

- Embedded systems & microcontroller programming.
- USB HID protocols.
- Serial communication and real-time feedback handling.
- Feedback & Error Handling Mechanisms
- Real-Time Systems & Timing Control

Latest Technologies

- ESP32-S3 with native USB OTG.
- ESP-IDF and TinyUSB HID library.
- Integration with AI-driven orchestration.(Agentic AI)

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Individual Title: HID Actuation

Evaluation & Validation

- Functional testing across Windows, Linux, and macOS.
- Latency: < 300 ms per action
- Accuracy: > 95% execution correctness.
- Reliability: > 85% task success in exploratory testing scenarios

Data & Ethical Clearance

- Data Privacy: The HID module does not store or transmit sensitive user data.
- Ethical Usage: Prevent misuse for malicious input injection by ensuring commands only come from authenticated sources.
- Safety & Compliance.
- Traceability.

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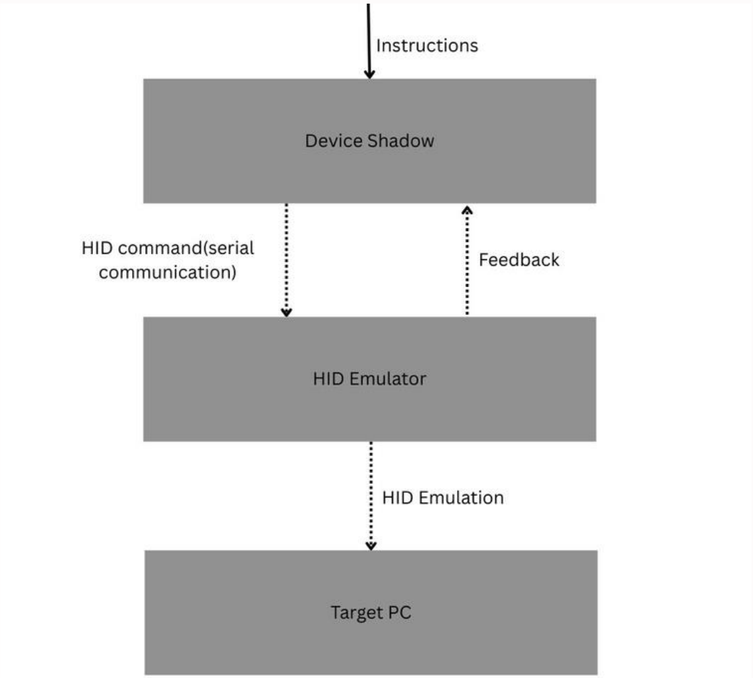
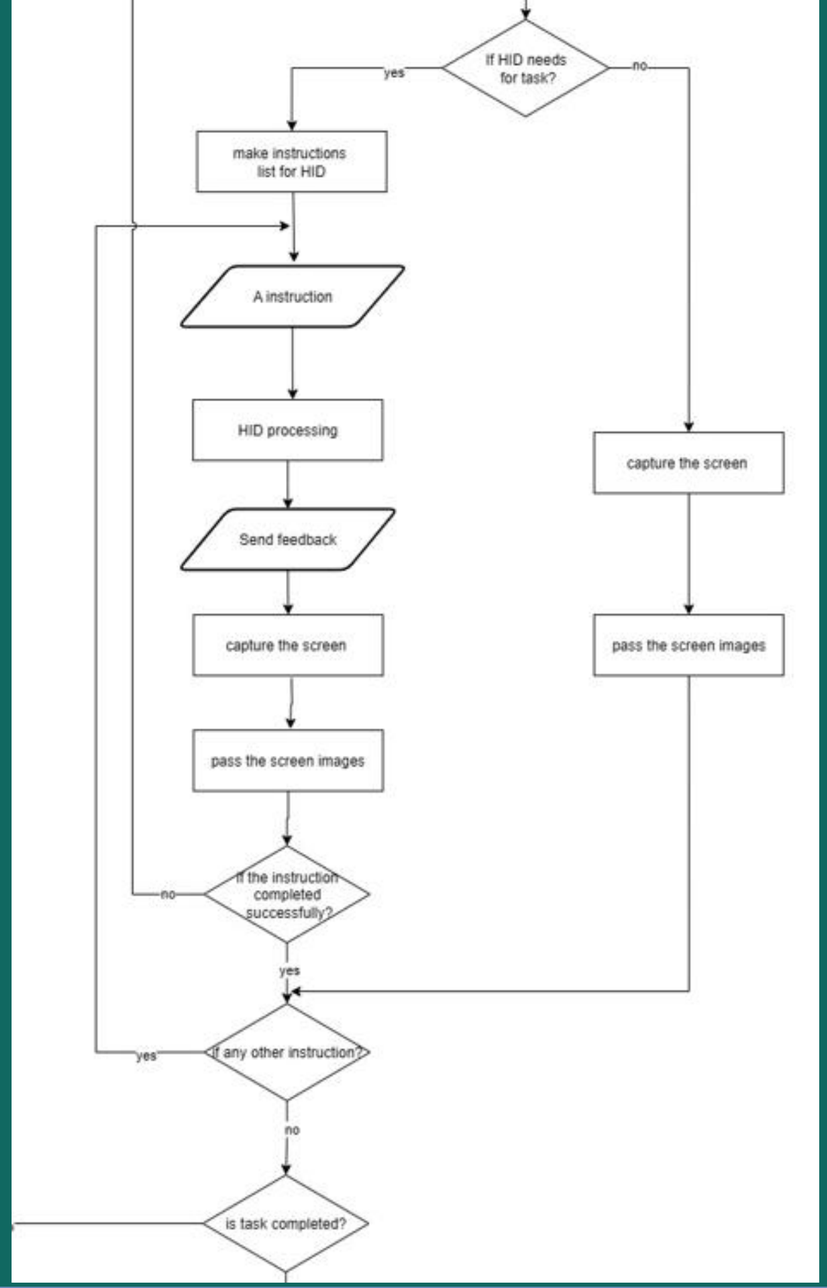
Individual Title: HID Actuation

System Overview





Real-World Usage Scenario

- *Instruction: “Press Enter”.*

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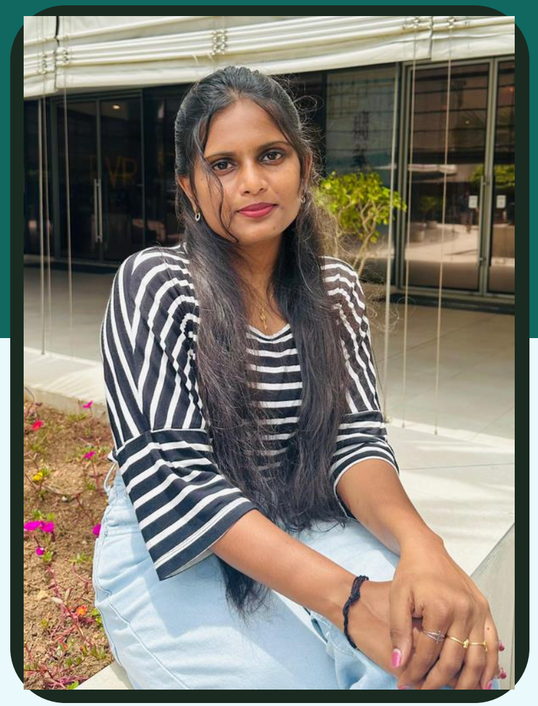
Title of the individual component: Agentic AI Logic

Specialization: Software Engineering



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IT22125248 | Annesiyani Srikanthan | 25-26J-477

Individual Title: Agentic AI Logic

💡 Proven Gap / Creative Solution

- Require internal access or APIs.[4][5]
- Can't adapt dynamically to unexpected UI changes.[7]
- **Creative solution:** Agentic AI acts as an intelligent orchestrator,
  - observing externally,
  - planning dynamically
  - adapting autonomously based on real-time feedback.

🔍 Knowledge Gap / Problem Definition

- Automation systems fail in black-box scenarios where internal data is inaccessible[4].
- Current tools lack adaptive decision-making, causing repeated failures with dynamic UIs.[5]

⚠️ Shortcomings of Existing Systems & ✅ Proposed Solution

Shortcomings	Proposed Solution
• Dependence on system APIs.[4]	• External perception with screen capture and HID control.
• Lack of adaptability[7]	• Real-time Perceive → Plan → Act → Evaluate loop.
• Weak reporting[4][5]	• Integrate logging and Jira for task tracking and reports.

IT22125248 | Annesiyani Srikanthan | 25-26J-477

Individual Title: Agentic AI Logic

Key Domains of Knowledge

- AI & Machine Learning → Adaptive Agentic AI
- NLP → Interpret LLM action plans
- Computer Vision → Detect UI changes
- Software Engineering → Orchestration & integration

Latest Technologies

- Agentic AI loop: Perceive → Plan → Act → Evaluate
- LLMs: GPT, LLaMA, Mistral
- Retries & recovery
- Jira API for task reporting & tracking

IT22125248 | Annesiyani Srikanthan | 25-26J-477

Individual Title: Agentic AI Logic

Evaluation & Validation

- Validation Approach: Run black-box automation test scenarios on real applications
- Key Metrics:
  - Task Success Rate (completion without human help)
  - Retry Frequency (efficiency of adaptive loop)
  - Decision Accuracy under dynamic UI changes
- Target Performance: ≥ 85% overall success rate across diverse test cases

Data & Ethical Clearance

- Data Sources: Screen captures, recordings, synthetic/dummy applications
- Privacy Protection: Not saving sensitive or personal data collected
- Ethical Assurance: External camera-based → non-intrusive, black-box safe
- Intended Use: Software testing & research purposes only (not for malicious activity)

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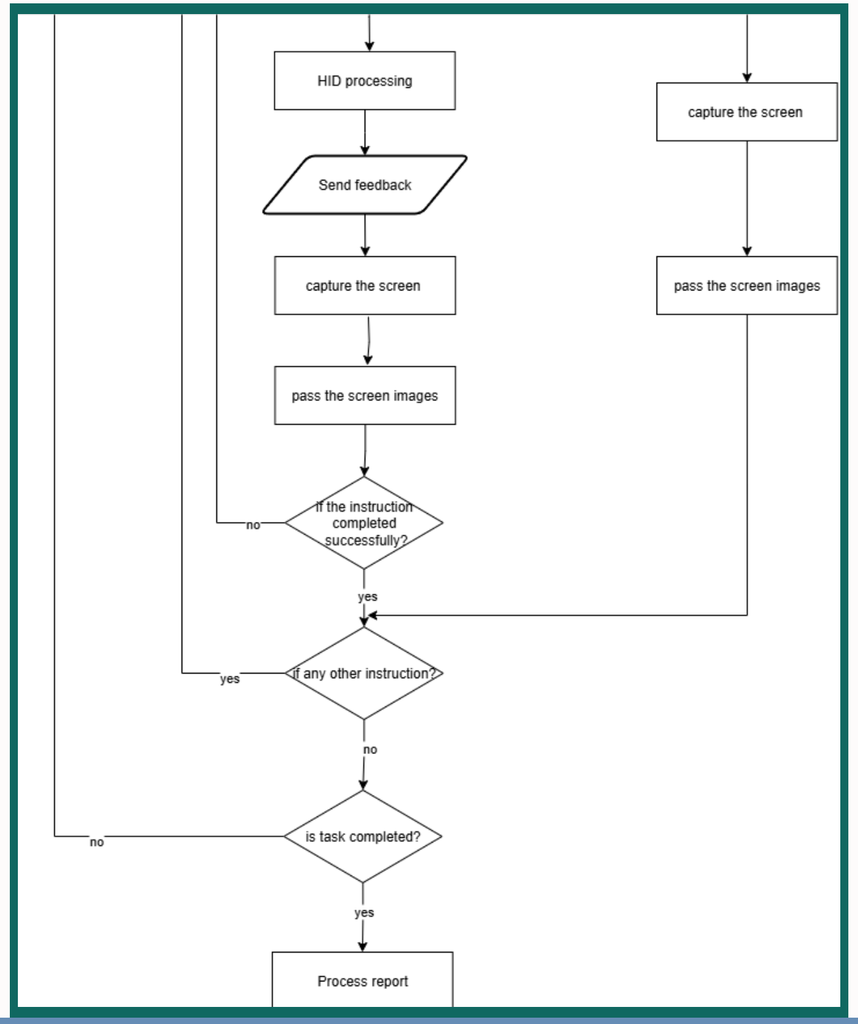
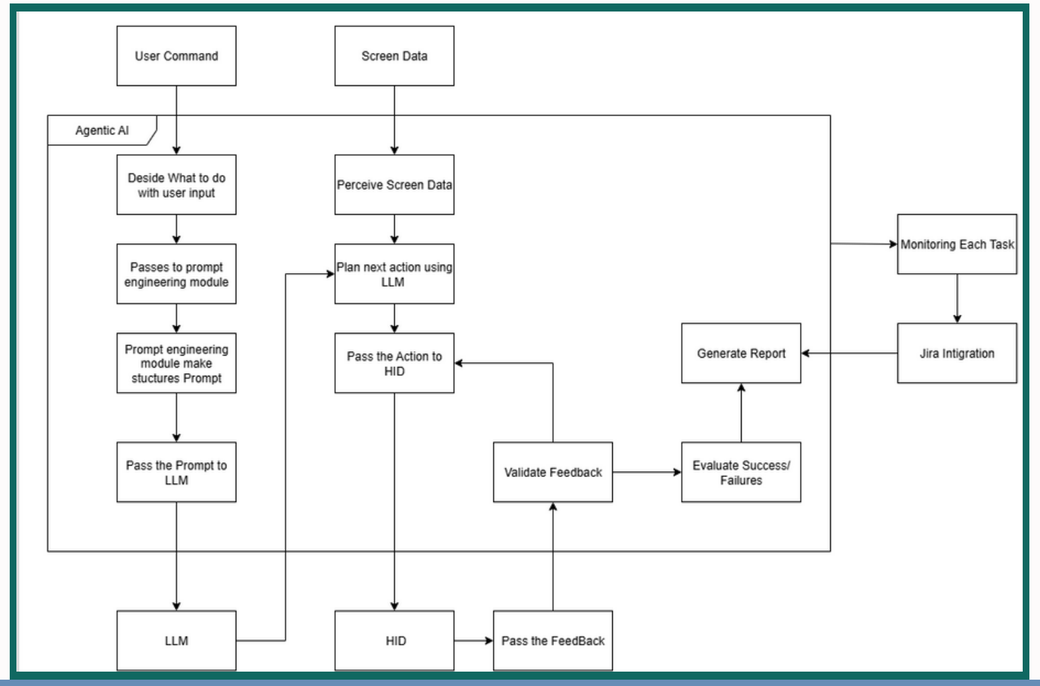
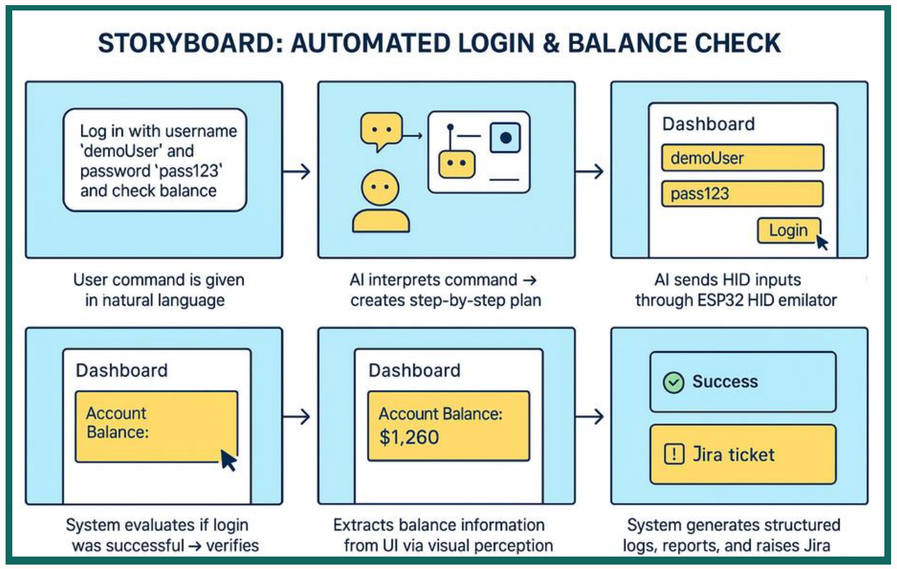
Individual Title: Agentic AI Logic

System Overview







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A Human-Like Agent For Screen-Aware Task Automation

Commercialization & Market Overview

Commercialization:

- Easy-to-use hardware/software for non-technical users
- Integrates with existing workflows
- Pilot tests and demos to attract clients

Total Investment: ~LKR 25,000

- ESP32 & Webcam: LKR 10,000
- Cloud GPU & Software: LKR 10,000
- Miscellaneous & Documentation: LKR 5,000

Cost Recovery:

- Sell 4 prototype kits (~LKR 8,000 each)
- Subscription: LKR 2,000–3,000/month
- Consultancy/pilot projects

Market & Value:

- SMEs, IT firms, educational labs
- Non-technical users benefit from human-like, scalable automation
- Protect IP through software copyright & automation patents

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[5] Appium, “Appium Official Documentation,” [Online]. Available: https://appium.io/docs/en/latest/. [Accessed: 31-Aug-2025].

[6] Google Developers, “UIAutomator Documentation,” [Online]. Available: https://developer.android.com/training/testing/ui-automator. [Accessed: 31-Aug-2025].

[7] UiPath

Survey Link - https://forms.gle/DZUU8bwG5NWPSGrV9

Thank You